# Short- and long-term temporal network prediction based on network memory

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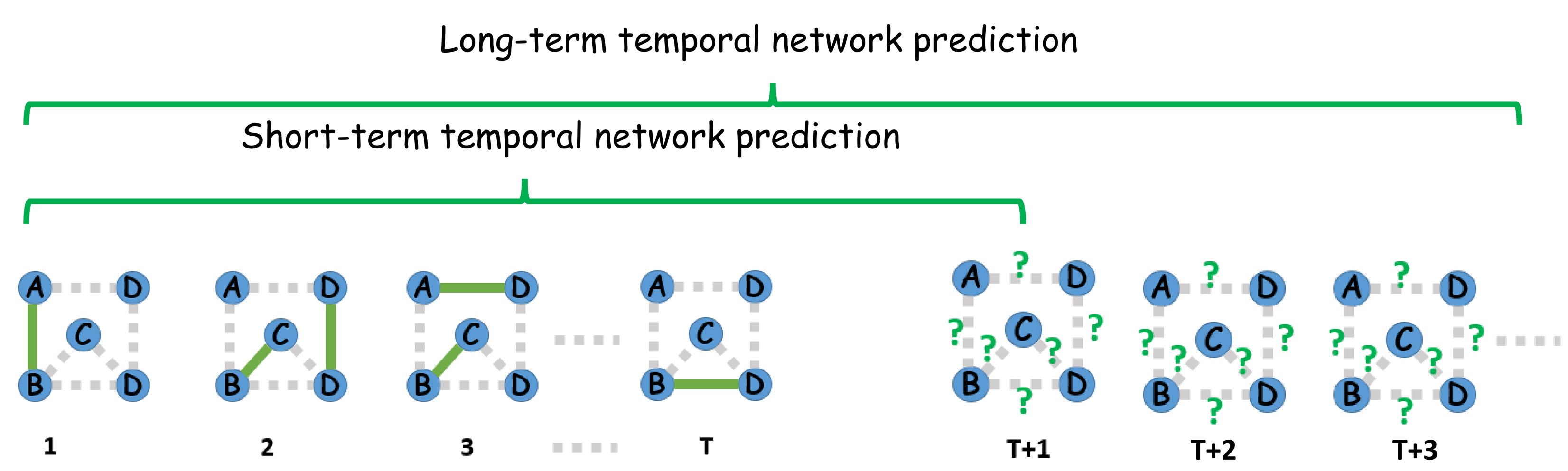
## Motivation & Research Goals

Temporal networks like physical contact networks are networks whose topology changes over time. Predicting future temporal networks is crucial e.g., to forecast and mitigate the spread of epidemics and misinformation on the network. The classic temporal network prediction problem that aims to predict the temporal network in the short-term future based on the network observed in the past has been addressed mostly via machine learning algorithms, at the expense of high computational costs and limited interpretation of the underlying mechanisms that form the networks. This motivates us to develop network-based models to predict future network based on the network properties of links observed in the past.

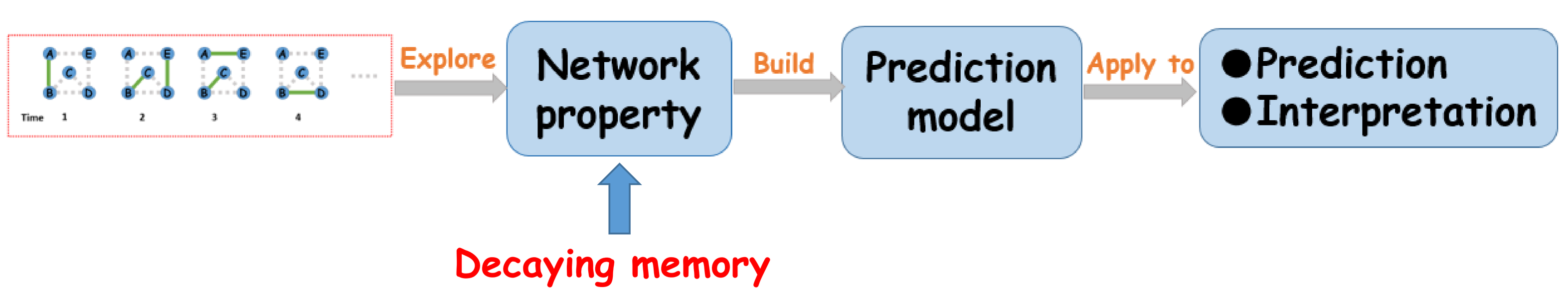
## Methods

(1) What is temporal network prediction?

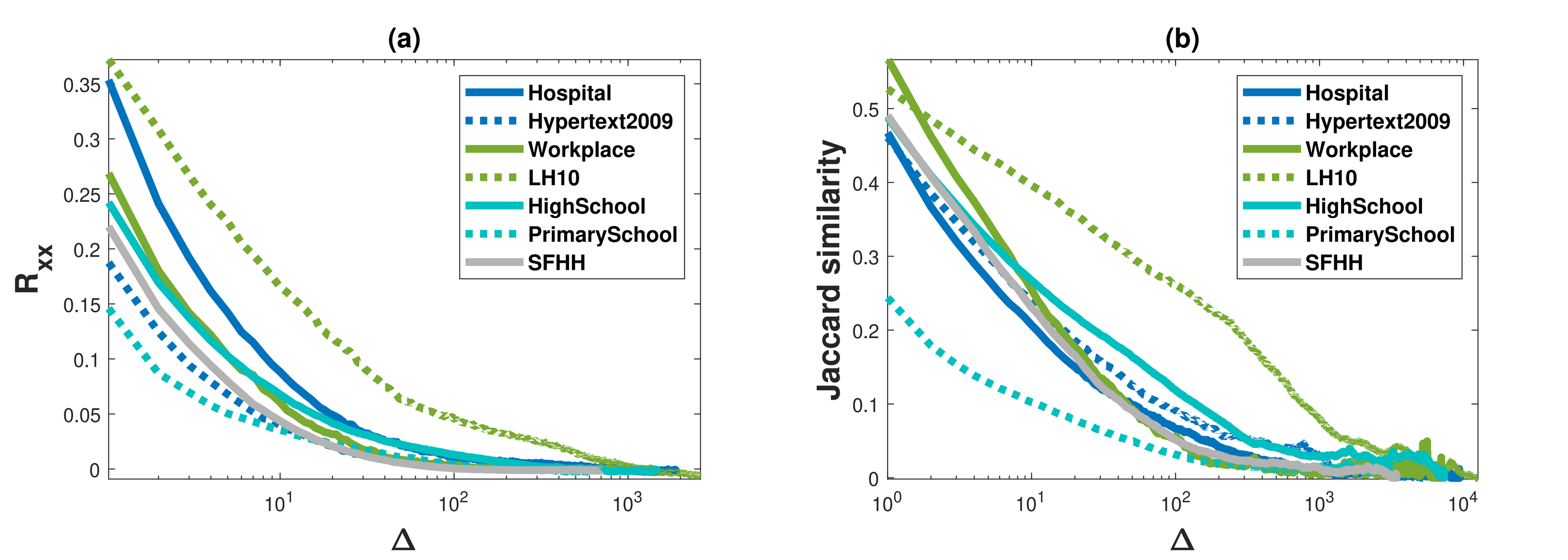



Short-term temporal network prediction is a task of predicting temporal network structure at future one-time step based on the network structure observed in previous steps. And long-term temporal network prediction is predicting temporal networks multi-time steps ahead based on the network topology observed in the past.

(2) How do we predict temporal networks?



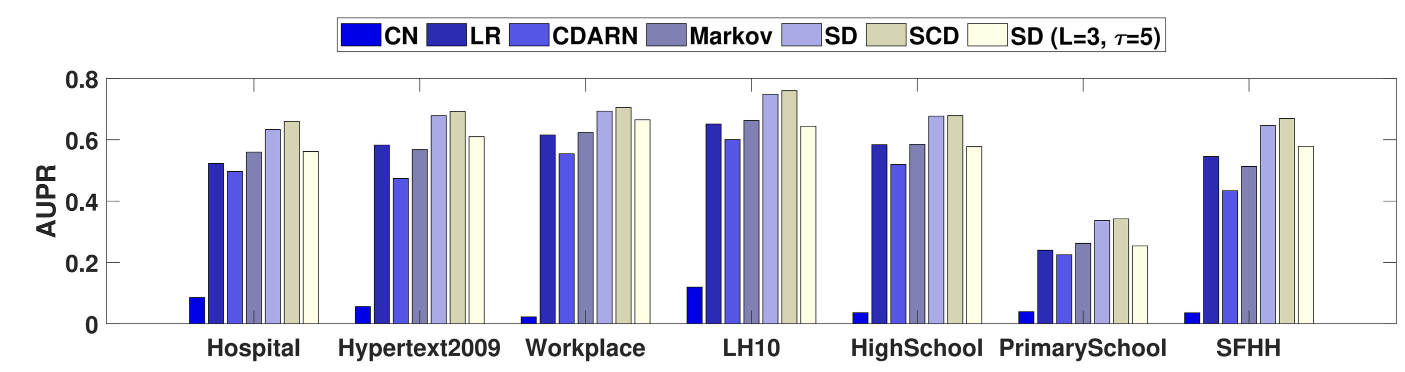

(3) What properties of temporal networks do we use?




We explore the similarity between the network topologies at any two-time steps with a given time lag $\Delta$. We find that the similarity is relatively high when the time lag is small and decreases as the time lag increases. Inspired by such time-decaying memory of temporal networks and recent advances, we propose two models that predict a link's activity at the next one-time step and multi-time steps ahead based on past activities of the link itself or also of the neighboring links, respectively.

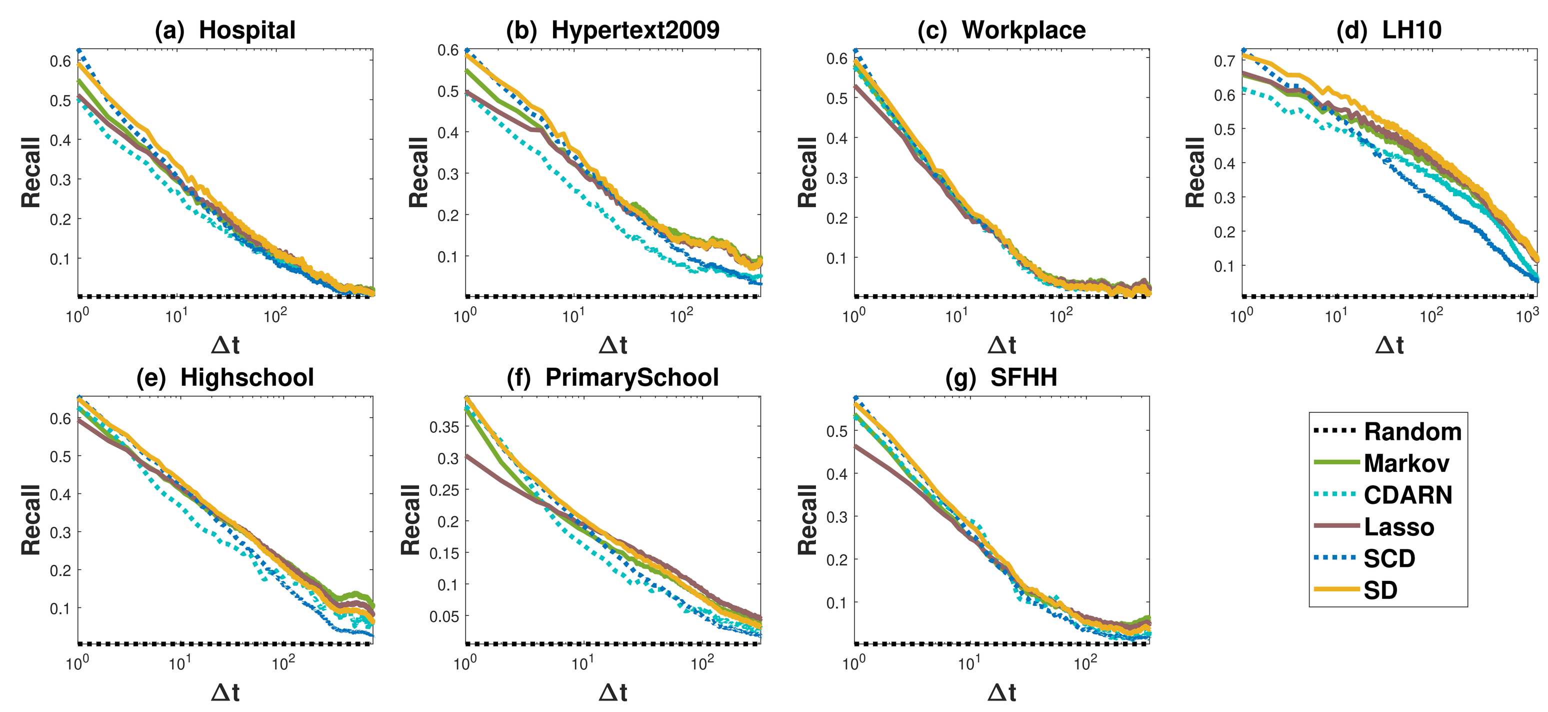
## Selected Results

(4) Short-term prediction results



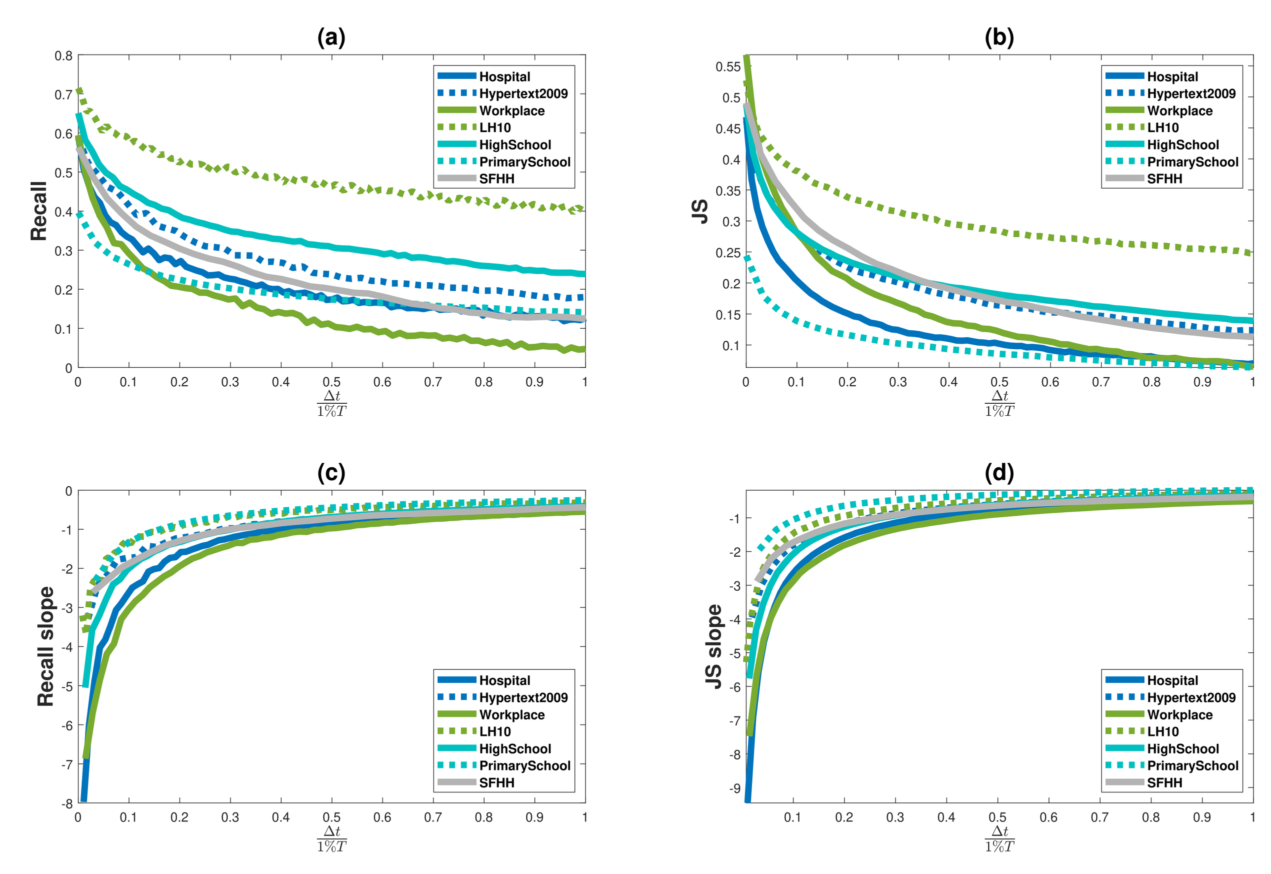

We find that our models (SD and SCD) outperform in both prediction quality (AUPR) and computational complexity, and predict better in networks that have a stronger memory.

(5) Long-term prediction results




Through exploring the prediction quality, Recall, for SD, SCD, Lasso Regression, CDARN, Markov, and Random model, respectively in seven temporal networks at each prediction step $t + \Delta t$, we find in general our SD model performs the best among all models in all data sets.

(6) Relation between the prediction quality of SD model and network memory




The prediction quality of our SD model decays as the prediction step is further ahead in time and this decay speed is positively correlated with the decay speed of network memory.

## Where you can find this work

Short- and long-term temporal network prediction based on network memory. L. Zou, A. Ceria, and H. Wang. 21 March 2023, PREPRINT (Version 1) available at Research Square [https://doi.org/10.21203/rs.3.rs-2702059/v1]